

Continuous assessment III

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The lorentze force is given by

$$F = q \left(\vec{E} + \vec{v} \times \vec{B} \right) \quad (1)$$

For circular motion we have

$$F = \frac{mv^2}{r} \quad (2)$$

As we are in circular motion we know $\vec{v} \perp \vec{B}$ and $\vec{E}$

$$\implies F = |q||\vec{v}| \cdot |\vec{B}| = \frac{mv^2}{r} \quad (3)$$

$$\implies |\vec{B}| = \frac{mv}{|q|r} = \frac{p}{|q|r} \quad (4)$$

We know the relativistic energy of a particle is given by

$$E = \sqrt{p^2 c^2 + m_0^2 c^4} \quad (5)$$

$$p^2 = \frac{E^2 - m_0^2 c^4}{c^2} \quad (6)$$

$$|\vec{B}| = \frac{\sqrt{E^2 - m_0^2 c^4}}{|q|rc} \quad (7)$$

$$(8)$$

For the electron we have

$$|\vec{B}_{elec}| = \frac{\sqrt{(27.5 \times 10^9 \cdot 1.6 \times 10^{-19})^2 - (9.1 \times 10^{-31})^2} (3 \times 10^8)^4}{1.6 \times 10^{-19} \cdot \frac{6.3 \times 10^3}{2\pi} \cdot 3 \times 10^8} \approx 9.14 \times 10^{-2} T \quad (9)$$

For the proton we have

$$|\vec{B}_{elec}| = \frac{\sqrt{(920 \times 10^9 \cdot 1.6 \times 10^{-19})^2 - (1.6726 \times 10^{-27})^2} (3 \times 10^8)^4}{1.6 \times 10^{-19} \cdot \frac{6.3 \times 10^3}{2\pi} \cdot 3 \times 10^8} \approx 3.06 T \quad (10)$$

(2)

$$\Delta E = \frac{2\pi}{\omega} P = \frac{2\pi r}{c\beta} \frac{2q^2 c \gamma^4 \beta^4}{3r^2} \quad (11)$$

$$\Delta E = \frac{4\pi q^2}{3r} \frac{\beta^3}{(1 - \beta^2)^2} \quad (12)$$

$$p^2 = \frac{E^2 - m_0^2 c^4}{c^2}, \implies p = \gamma m_0 v = \frac{\sqrt{E^2 - m_0^2 c^4}}{c} \quad (13)$$

$$\beta^2 = (1 - \beta^2) \left(\frac{E^2}{m_0^2 c^4} - 1 \right) = \frac{E^2}{m_0^2 c^4} - 1 - \frac{E^2}{m_0^2 c^4} \beta^2 + \beta^2 \quad (14)$$

$$\beta^2 = 1 - \frac{m_0^2 c^4}{E^2} \quad (15)$$

Thus

$$\Delta E = \frac{4\pi q^2}{3r} \frac{\left(1 - \frac{m_0^2 c^4}{E^2}\right)^{\frac{3}{2}}}{\left(\frac{m_0^2 c^4}{E^2}\right)^2} = \frac{4\pi q^2}{3r} \left(1 - \frac{m_0^2 c^4}{E^2}\right)^{\frac{3}{2}} \frac{E^4}{m_0^4 c^8} \quad (16)$$

For an electron we have

$$\Delta E_{el} = \frac{4\pi (1.6 \times 10^{-19})^2}{3 \left(\frac{6.3 \times 10^3}{2\pi} \right)} \left(1 - \frac{(9.1 \times 10^{-31})^2 (3 \times 10^8)^4}{(27.5 \times 10^9 \cdot 1.6 \times 10^{-19})^2} \right)^{\frac{3}{2}} \frac{(27.5 \times 10^9 \cdot 1.6 \times 10^{-19})^4}{(9.1 \times 10^{-31})^4 (3 \times 10^8)^8} = 8.91 \times 10^{-22} J = 5.57 meV \quad (17)$$

For the proton we have

$$\Delta E_p = \frac{4\pi (1.6 \times 10^{-19})^2}{3 \left(\frac{6.3 \times 10^3}{2\pi} \right)} \left(1 - \frac{(1.6726 \times 10^{-27})^2 (3 \times 10^8)^4}{(920 \times 10^9 \cdot 1.6 \times 10^{-19})^2} \right)^{\frac{3}{2}} \frac{(920 \times 10^9 \cdot 1.6 \times 10^{-19})^4}{(1.6726 \times 10^{-27})^4 (3 \times 10^8)^8} = 9.78 \times 10^{-29} J = 611 peV \quad (18)$$

(3)

$$\frac{\dot{p}}{m_0 c^2} = \dot{\gamma} \beta + \gamma \dot{\beta} \quad (19)$$

$$\dot{\gamma} = -\frac{1}{2} (1 - \beta^2)^{-\frac{3}{2}} (-2\beta \cdot \dot{\beta}) = \beta \cdot \dot{\beta} \gamma^3 \quad (20)$$

$$\Rightarrow \frac{\dot{p}}{m_0 c} = \beta \cdot \dot{\beta} \gamma^3 \beta + \gamma \dot{\beta} = \gamma (|\beta| |\dot{\beta}| \gamma^2 + \dot{\beta}) \quad (21)$$

So

$$\frac{\dot{p}^2}{m_0^2 c^4 \gamma^2} = (|\beta|^2 |\dot{\beta}|^2 \gamma^4 + \dot{\beta}^2 + 2|\dot{\beta}| \beta \cdot \dot{\beta}) \quad (22)$$

$$= (1 + \gamma^2 \beta^2)^2 \dot{\beta}^2 = \gamma^4 \dot{\beta}^2 \Rightarrow \dot{\beta}^2 = \frac{\dot{p}^2}{\gamma^6 m_0^2 c^2} \quad (23)$$

So we get

$$P_{lin} = \frac{2q^2 \gamma^6}{3c} \frac{\dot{p}^2}{\gamma^6 m_0^2 c^2} \Big|_{ret} = \frac{2q^2}{3m_0^2 c^3} \dot{p}^2 \Big|_{ret} \quad (24)$$

For the circular motion case we have the Larmor equation

$$P(t) = \frac{2q^2}{3c} \left[\gamma^6 \left[\dot{\beta}^2 - (\beta \times \dot{\beta})^2 \right] \right]_{ret} \quad (25)$$

We have for the circular case that $\beta \perp \dot{\beta}$.

$$P(t)_c = \frac{2q^2}{3c} \left[\gamma^6 \left[\dot{\beta}^2 - (|\beta| |\dot{\beta}|)^2 \right] \right]_{ret} = \frac{2q^2}{3c} \left[\gamma^6 \left[\dot{\beta}^2 - (|\beta| |\dot{\beta}|)^2 \right] \right]_{ret} = \frac{2q^2}{3c} \left[\gamma^6 \left[\dot{\beta}^2 - (|\beta| |\dot{\beta}|)^2 \right] \right]_{ret} \quad (26)$$

$$P(t)_c = \frac{2q^2}{3c} \left[\gamma^6 \dot{\beta}^2 [1 - (|\beta|)^2] \right]_{ret} = \frac{2q^2}{3c} \left[\gamma^4 \dot{\beta}^2 \right]_{ret} \quad (27)$$

$$\Rightarrow P(t)_c = \frac{2q^2 \gamma^4}{3c} \frac{\dot{p}^2}{\gamma^2 m_0^2 c^2} \Big|_{ret} = \frac{2q^2}{3m_0^2 c^3} \gamma^2 \dot{p}^2 \Big|_{ret} \quad (28)$$

$$(29)$$

We can notice that

$$P(t)_c = \gamma^2 \Big|_{ret} P_{lin} \quad (30)$$

So we see that the power radiated is larger for the circular case as $\gamma > 1$. It is more efficient to build a linear accelerator as there is less radiation loss.

Question 2

0.0.1 (1)

(a)

Given

$$\Re B = \vec{b}_0 \sin(kz - \omega t) \quad (31)$$

where we know that the electric field and the magnetic field are orthogonal to the direction of propagation $\hat{z}$ (32)

in this case. Thus they lie in the x-y plane (33)

$$\vec{b}_0 = (b_{0x}\hat{x}, b_{0y}\hat{y}, 0\hat{z}) \quad \vec{e}_0 = (e_{0x}\hat{x}, e_{0y}\hat{y}, 0\hat{z}) \quad (34)$$

$$\vec{\nabla} \times \vec{E} = \frac{\partial \vec{B}}{\partial t} = -\omega \vec{b}_0 \cos(kz - \omega t) \quad (35)$$

Let the Electric field be of the form

$$\vec{E}(z, t) = \vec{e}_0 \sin(kz - \omega t) \quad \implies \vec{\nabla} \times \vec{E} = k(\hat{z} \times \vec{e}_0) \cos(kz - \omega t) \quad (36)$$

We can compare coefficients to get

$$-\omega \vec{b}_0 = k(\hat{z} \times \vec{e}_0) \quad \text{with} \quad \omega = kc \quad (37)$$

$$\hat{z} \times \vec{e}_0 = c \vec{b}_0 \underbrace{\implies}_{\vec{e}_0 \perp \hat{z}} \vec{e}_0 = c \hat{z} \times \vec{b}_0 \quad (38)$$

So our electric field is of the form

$$\Re \vec{E} = -c(-b_{0y}\hat{x} + b_{0x}\hat{y}) \sin(kz - \omega t) \quad (39)$$

$$(40)$$

(b)

When we are given

$$\Re B = b_0(\hat{x} \cos(kz - \omega t) + \hat{y} \sin(kz - \omega t)) \quad (41)$$

$$\vec{\nabla} \times \vec{E} = \frac{\partial \vec{B}}{\partial t} = b_0(\omega \hat{x} \sin(kz - \omega t) - \omega \cos(kz - \omega t) \hat{y}) \quad (42)$$

$$\vec{\nabla} \times \Re \vec{E} = e_{0i} \begin{vmatrix} \partial_x & \partial_y & \partial_z \\ \sin(kz - \omega t) & \cos(kz - \omega t) & 0 \end{vmatrix} = e_{0i}(k \sin(kz - \omega t) + k \cos(kz - \omega t) \hat{y}) \quad \text{with } i \in \{x, y\} \quad (43)$$

$$\implies ke_{0i} = b_0\omega \quad \implies \frac{b_0\omega}{k} = e_{0i} \quad \text{we can see a minus difference in } \hat{x} \text{ thus we get} \quad (44)$$

our electric field is of the form

$$\vec{E} = \frac{b_0\omega}{k}(\sin(kz - \omega t) \hat{x} - \cos(kz - \omega t) \hat{y}) \quad (45)$$

$$\vec{E} = b_0c(\sin(kz - \omega t) \hat{x} - \cos(kz - \omega t) \hat{y}) \quad (46)$$

$$(47)$$

(2)

We have complex fields given by

$$\vec{E}(z, t) = \left[\vec{E}_0 e^{i(kz - \omega t)} \right] \quad \vec{B}(z, t) = \left[\vec{B}_0 e^{i(kz - \omega t)} \right] \quad (48)$$

$$(49)$$

(a)

For the case with

$$\Re B = b_0 \sin(kz - \omega t) = \Re \left[\vec{B}_0 e^{i(kz - \omega t)} \right] = \Re \left[\vec{B}_0 (\cos(kz - \omega t) + i \sin(kz - \omega t)) \right] \quad (50)$$

$$(51)$$

We see we need it to match the sine term so we can let $\vec{B}_0 = -ib_0$

$$\implies \vec{B}(z, t) = -ib_0 e^{i(kz - \omega t)} \quad (52)$$

We had the electric field given by

$$\Re \vec{E} = -c (-b_{0y} \hat{x} + b_{0x} \hat{y}) \sin(kz - \omega t) \quad (53)$$

$$(54)$$

So the complex field is given by

$$\vec{E} = ci (-b_{0y} \hat{x} + b_{0x} \hat{y}) e^{i(kz - \omega t)} \quad (55)$$

(b)

For the second case with

$$\Re B = b_0 (\hat{x} \cos(kz - \omega t) + \hat{y} \sin(kz - \omega t)) = \Re \left[\vec{B}_0 e^{i(kz - \omega t)} \right] = \Re \left[\vec{B}_0 (\cos(kz - \omega t) + i \sin(kz - \omega t)) \right] \quad (56)$$

$$(57)$$

We can see that we need to make the y component positive and real while the x component fits, thus

$$\vec{B}_0 = b_0 (\hat{x} - i \hat{y}) \quad (58)$$

$$\implies \vec{B}(z, t) = b_0 (\hat{x} - i \hat{y}) e^{i(kz - \omega t)} \quad (59)$$

We had the electric field given by

$$\vec{E} = b_0 c (\sin(kz - \omega t) \hat{x} - \cos(kz - \omega t) \hat{y}) \quad (60)$$

$$(61)$$

So the Complex electric field is given by

$$\vec{E} = -cb_0 e^{i(kz - \omega t)} (i \hat{x} + \hat{y}) \quad (62)$$

(3)

The pointing vector is given by

$$\vec{S} = \frac{1}{\mu} \vec{E} \times \vec{B} \quad (63)$$

Here as we only want the observable section of the pointing vector we are only interested in the real part of our vectors.

$$\vec{S} = \frac{1}{\mu} \Re \vec{E} \times \Re \vec{B} \quad (64)$$

(a)

$$\vec{S} = \frac{1}{\mu} \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ b_{0x} \sin(kz - \omega t) & b_{0y} \sin(kz - \omega t) & 0 \\ -cb_{0y} \sin(kz - \omega t) & cb_{0x} \sin(kz - \omega t) & 0 \end{vmatrix} = \left(0\hat{i}, 0\hat{j}, -cb_0^2 \frac{\sin^2(kz - \omega t)}{\mu} \hat{k} \right) \quad (65)$$

$$\langle \vec{S} \rangle = \frac{1}{T} \int_0^T \vec{S} dt = \frac{\omega}{2\pi} \int_0^{\frac{\pi}{\omega}} cb_0^2 \frac{\sin^2(kz - \omega t)}{\mu} \hat{k} dt \quad (66)$$

$$\langle \vec{S} \rangle = \frac{-\omega cb_0^2}{4\pi\mu} \int_0^{\frac{2\pi}{\omega}} 1 - \cos(2kz - 2\omega t) dt \hat{k} = \frac{\omega cb_0^2}{4\pi\mu} \left[\frac{2\pi}{\omega} - \frac{\sin(2kz - 4\pi)}{2\omega} + \frac{\sin(2kz)}{2\omega} \right] \quad (67)$$

$$\langle \vec{S} \rangle = \frac{cb_0^2}{2\mu} \hat{k} \quad (68)$$

(b)

$$\vec{S} = -\frac{1}{\mu} \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ -b_0 \sin(kz - \omega t) & cb_0 \cos(kz - \omega t) & 0 \\ b_0 \cos(kz - \omega t) & cb_0 \sin(kz - \omega t) & 0 \end{vmatrix} = \left(0\hat{i}, cb_0^2 - (\sin(kz - \omega t)^2 + \cos(kz - \omega t)^2) \hat{j}, \right) \quad (69)$$

$$\vec{S} = \frac{cb_0^2}{\mu} \hat{k} \quad (70)$$

We have a constant so we know the time average of a constant will be the constant

$$\langle \vec{S} \rangle = \vec{S} = \frac{cb_0^2}{\mu} \hat{k} \quad (71)$$

The shortcut method

$$\vec{S} = \frac{1}{2\mu} \Re \left(\vec{E} \times \vec{B}^* \right) \quad (72)$$

$$(73)$$

(a)

$$\vec{S} = \frac{1}{2\mu} \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ +cib_{0y}e^{i(kz-\omega t)} & -cib_{0x}e^{i(kz-\omega t)} & 0 \\ ib_{0x}e^{-i(kz-\omega t)} & ib_{0y}e^{-i(kz-\omega t)} & 0 \end{vmatrix} = \frac{cb_{0y}^2 + b_{0x}^2 c}{2\mu} \hat{k} \quad (74)$$

(b)

$$\vec{S} = \frac{1}{2\mu} \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ cb_0ie^{i(kz-\omega t)} & cb_0e^{i(kz-\omega t)} & 0 \\ b_0e^{-i(kz-\omega t)} & ib_0e^{-i(kz-\omega t)} & 0 \end{vmatrix} = \frac{cb_0^2}{\mu} \hat{k} \quad (75)$$

(4)

The radiation pressure is given by

$$P = \frac{F}{A} = \frac{\vec{S}}{v} \quad (76)$$

In this case $v = c$ α is the angle from the z direction to the plane and we have the force given by

$$F = \frac{\langle S \rangle \sin \alpha}{c} \hat{z} \quad (77)$$

$$\Rightarrow P = \frac{\langle S \rangle \sin \alpha}{c} \quad (78)$$

For the first case

(a)

$$\langle S \rangle = \frac{c b_0^2}{2\mu} \quad (79)$$

$$\Rightarrow P = \frac{b_0^2 \sin \alpha}{2\mu} \quad (80)$$

For the second case

(b)

$$\langle S \rangle = \frac{c b_0^2}{\mu} \quad (81)$$

$$\Rightarrow P = \frac{b_0^2 \sin \alpha}{\mu} \quad (82)$$